ACADEMIC NOTES

Unit I

Functions & Mathematical Foundations

Sets · Function Types · Graphs & Transformations · Limits & Continuity

Applications: Activation Functions · Growth Models in AI

Subject:	Mathematics for Artificial Intelligence / Machine Learning
Unit:	Unit I — Functions and Mathematical Foundations
Topics:	Sets, Linear, Polynomial, Exponential, Logarithmic Functions, Graphs, Transformations, Limits, Continuity
Level:	Undergraduate / Postgraduate

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1. Sets and Functions

1.1 Sets — Fundamentals

A **set** is a well-defined collection of distinct objects called **elements**. Sets form the foundation of all mathematical structures.

Set Notation	A = {1, 2, 3} — listing elements. A = {x | x > 0} — set-builder form.
Element (∈)	a ∈ A means "a is an element of set A". a ∉ A means a is not an element.
Subset (⊆)	A ⊆ B if every element of A is also in B. A ⊂ B for proper subset.
Union (∪)	A ∪ B = {x | x ∈ A or x ∈ B} — all elements in either set.
Intersection (∩)	A ∩ B = {x | x ∈ A and x ∈ B} — elements common to both sets.
Complement (A■)	A■ = {x ∈ U | x ∉ A} — all elements not in A, within universe U.
Cartesian Product	A × B = {(a, b) | a ∈ A, b ∈ B} — all ordered pairs.

1.2 Functions — Definition

A **function** f: A → B is a rule that assigns to each element x ∈ A (domain) exactly one element f(x) ∈ B (codomain). The set of all output values is the **range**.

Function notation: f: A → B, y = f(x)
Domain: set of all valid inputs (x values)
Range/Image: set of all output values f(x)
Codomain: set in which outputs live (range ⊆ codomain)

1.3 Properties of Functions

Property	Definition	Test
Injective (One-to-One)	f(a)=f(b) ■ a=b No two inputs give same output	Horizontal Line Test (every line hits ≤1 point)
Surjective (Onto)	Every b ∈ B has some a with f(a)=b Range = Codomain	Every output value is achieved
Bijective	Both injective and surjective Perfect pairing	Invertible function f■¹ exists
Composition	(g■f)(x) = g(f(x)) Apply f first, then g	Domain of g■f = domain of f where f(x) ∈ domain of g

2. Types of Functions

Functions are classified by their algebraic form and behaviour. Understanding each type is essential for modelling real-world phenomena and AI systems.

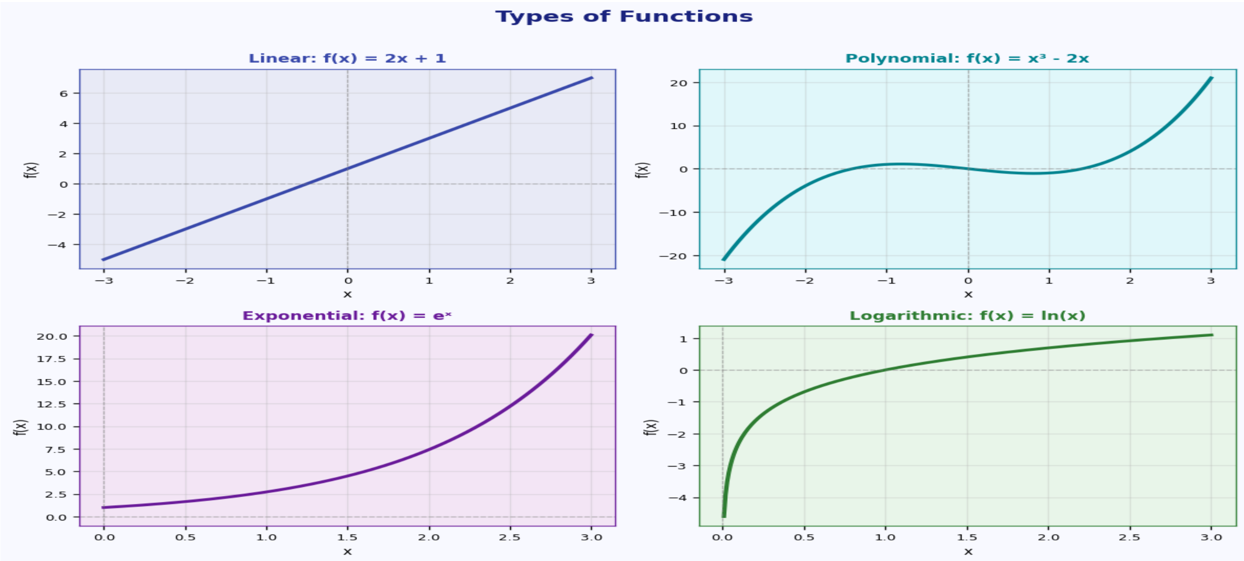

Figure: Visual overview of the four core function types

2.1 Linear Functions

f(x) = mx + b

m = slope (rate of change), b = y-intercept

Slope formula: m = (y2 - y1) / (x2 - x1)

Key Properties:

- Graph is always a straight line.
- Constant rate of change (slope m is fixed).
- If m > 0: increasing; m < 0: decreasing; m = 0: horizontal (constant).
- Domain and range: all real numbers ■ (unless restricted).

Special Forms:

Form	Equation	Use
Slope-Intercept	y = mx + b	Direct reading of slope & intercept
Point-Slope	y - y■ = m(x - x■)	Given a point and slope
Standard Form	Ax + By = C	Integer coefficients
Two-Point Form	y-y■ = [(y■-y■)/(x■-x■)](x-x■)	Given two points

2.2 Polynomial Functions

$f(x) = a_n \cdot x^n + a_{(n-1)} \cdot x^{(n-1)} + \ldots + a_1 \cdot x + a_0$

n = degree (highest power), $a_n \neq 0$ = leading coefficient

Common Polynomial Types by Degree:

Degree	Name	Form	Shape
Degree 0	Constant	$f(x) = c$	Horizontal line
Degree 1	Linear	$f(x) = mx + b$	Straight line
Degree 2	Quadratic	$f(x) = ax^2+bx+c$	Parabola (U-shape)
Degree 3	Cubic	$f(x) = ax^3+bx^2+cx+d$	S-shaped curve
Degree 4	Quartic	f(x) = ax■+...	W or M shape
Degree n	nth-degree	f(x) = a_n·x■+...	n roots (max)

Key Theorems:

- **Fundamental Theorem of Algebra:** A degree-n polynomial has exactly n roots (counting multiplicity) in ■.
- **Quadratic Formula:** $x = [-b \pm \sqrt{(b^2-4ac)}] / 2a$ for $f(x) = ax^2+bx+c$.
- **Remainder Theorem:** When $f(x)$ is divided by $(x-c)$, the remainder is $f(c)$.

2.3 Exponential Functions

f(x) = a · b^x (b > 0, b ≠ 1, a ≠ 0)
Natural exponential: f(x) = e^x where e ≈ 2.71828
Growth (b > 1): population, compound interest, AI model scaling
Decay (0 < b < 1): radioactive decay, cooling, dropout in NN

Properties of Exponential Functions:

- Domain: all real numbers ■. Range: (0, ∞) for a > 0.
- Horizontal asymptote at y = 0 (the x-axis).
- Always positive; never crosses x-axis.
- Passes through (0, a) since b■ = 1.

Laws of Exponents:

b■ · b■ = b■■■	**b■ / b■ = b■■■**
(b■)■ = b■■	**(ab)■ = a■ · b■**
b■ = 1	**b■■ = 1/b■**

2.4 Logarithmic Functions

f(x) = log_b(x) is the inverse of g(x) = b^x
log_b(x) = y ■ b^y = x (b > 0, b ≠ 1, x > 0)
Common log: log■■(x) = log(x) Natural log: log_e(x) = ln(x)

Properties of Logarithms (Log Laws):

log_b(MN) = log_b(M) + log_b(N)	Product Rule
log_b(M/N) = log_b(M) - log_b(N)	Quotient Rule
log_b(M■) = p · log_b(M)	Power Rule
log_b(b) = 1	Identity
log_b(1) = 0	Zero Property
log_b(M) = ln(M)/ln(b) = log(M)/log(b)	Change of Base

■ ***Note:*** *log_b(x) is only defined for x > 0. Domain: (0, ∞), Range: ■.*

3. Graphs and Transformations

Transformations allow us to obtain the graph of a related function from a known base graph, without recomputing from scratch. Every transformation has a geometric interpretation.

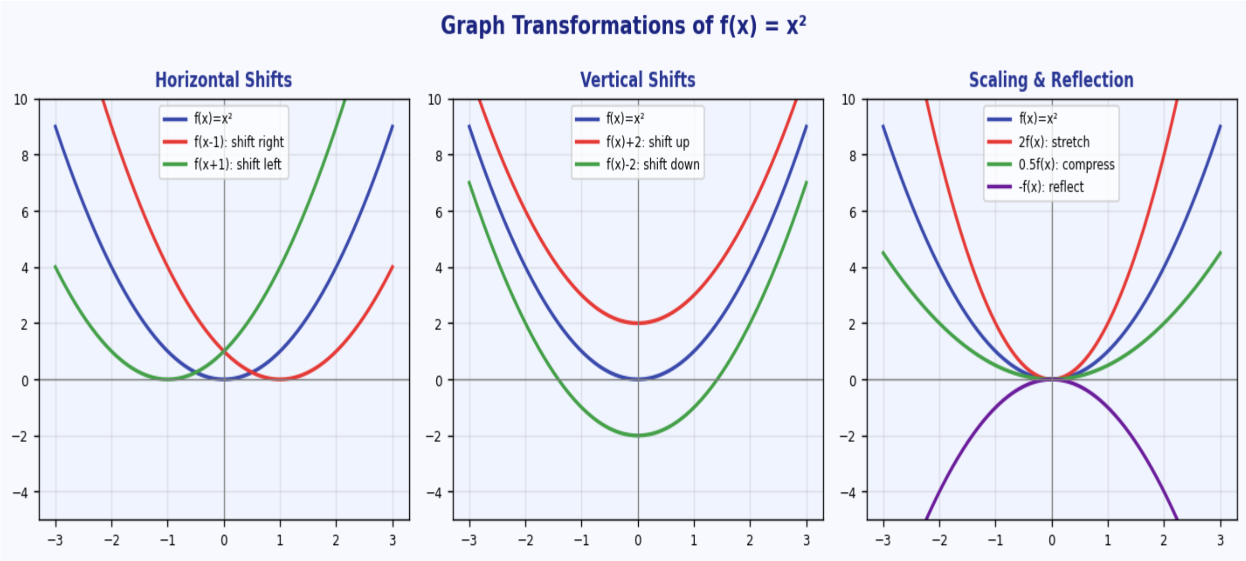

Figure: Horizontal Shifts, Vertical Shifts, Scaling & Reflection of $f(x) = x^2$

Transformation	Equation Change	Effect on Graph	Example
Vertical Shift ↑	$f(x) + k$ $(k > 0)$	Move graph UP by k units	$x^2 + 3$
Vertical Shift ↓	$f(x) - k$ $(k > 0)$	Move graph DOWN by k units	$x^2 - 2$
Horizontal Shift →	$f(x - h)$ $(h > 0)$	Move graph RIGHT by h units	$(x-1)^2$
Horizontal Shift ←	$f(x + h)$ $(h > 0)$	Move graph LEFT by h units	$(x+2)^2$
Vertical Stretch	$a\cdot f(x)$ $(a > 1)$	Stretch away from x-axis	$3x^2$
Vertical Compress	$a\cdot f(x)$ $(0<a<1)$	Compress toward x-axis	$0.5x^2$
Reflect over x-axis	$-f(x)$	Flip vertically	$-x^2$
Reflect over y-axis	$f(-x)$	Flip horizontally	$(-x)^2=x^2$
Horizontal Stretch	$f(b\cdot x)$ $(0<b<1)$	Stretch away from y-axis	$f(0.5x)$
Horizontal Compress	$f(b\cdot x)$ $(b > 1)$	Compress toward y-axis	$f(2x)$

General form: $y = a \cdot f(b(x - h)) + k$

a: vertical scale/reflection | b: horizontal scale/reflection

h: horizontal shift | k: vertical shift

Order: 1) Horizontal shift, 2) Horizontal scale, 3) Vertical scale, 4) Vertical shift

4. Limits and Continuity

Limits are the foundation of **calculus**. They describe the behaviour of a function as the input approaches a particular value, even if the function is not defined there.

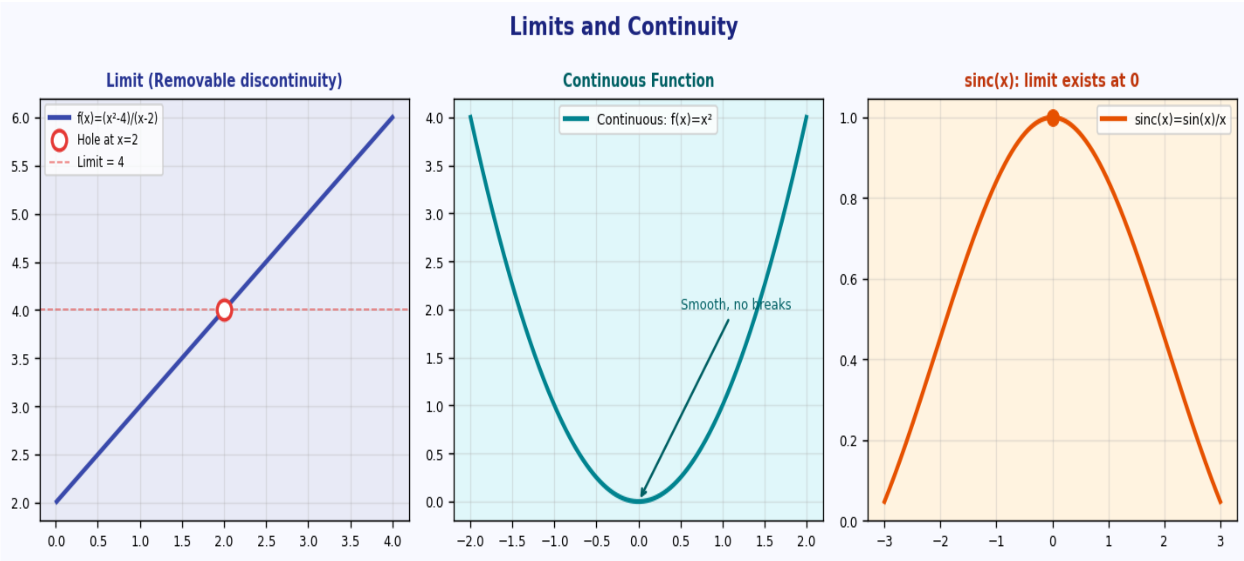

Figure: Limit approaching a value, continuous function, and sinc(x)

4.1 Definition of a Limit

lim_{x→a} f(x) = L

"As x approaches a, f(x) approaches L"

This does NOT require f(a) to be defined!

For the limit to exist, the **left-hand limit** and **right-hand limit** must both exist and be equal:

lim_{x→a■} f(x) = lim_{x→a■} f(x) = L ■ lim_{x→a} f(x) = L

4.2 Limit Laws

Sum	lim[f(x) + g(x)] = lim f(x) + lim g(x)
Difference	lim[f(x) - g(x)] = lim f(x) - lim g(x)
Product	lim[f(x)·g(x)] = lim f(x) · lim g(x)
Quotient	lim[f(x)/g(x)] = lim f(x) / lim g(x), if lim g(x) ≠ 0
Constant	lim[c·f(x)] = c · lim f(x)
Power	lim[f(x)■] = [lim f(x)]■
Squeeze	If g(x) ≤ f(x) ≤ h(x) and lim g = lim h = L, then lim f = L

4.3 Continuity

A function $f(x)$ is **continuous at $x = a$** if all three conditions hold:

1. $f(a)$ is defined (no holes or undefined points)
2. $\lim_{x \to a} f(x)$ exists (left & right limits are equal)
3. **$\lim_{x \to a} f(x) = f(a)$ (limit equals the function value)**

Types of Discontinuity:

Type	Condition	Example
Removable (Hole)	lim exists but $f(a)$ is undefined or $\lim \neq f(a)$	$f(x) = (x^2-4)/(x-2)$ at $x=2$ Limit = 4, but $f(2)$ undefined
Jump	Left & right limits exist but are unequal	Piecewise: $f(x)=1$ $(x<0)$, $f(x)=2$ $(x\geq 0)$
Infinite	$\lim = \pm\infty$ at some point	$f(x) = 1/x$ at $x = 0$
Oscillating	Function oscillates infinitely near the point	$f(x) = \sin(1/x)$ at $x = 0$

5. AI Applications: Activation Functions

Activation functions are the mathematical functions applied at each neuron in a neural network. They introduce **non-linearity**, enabling networks to learn complex patterns. They are drawn directly from the function types studied in this unit.

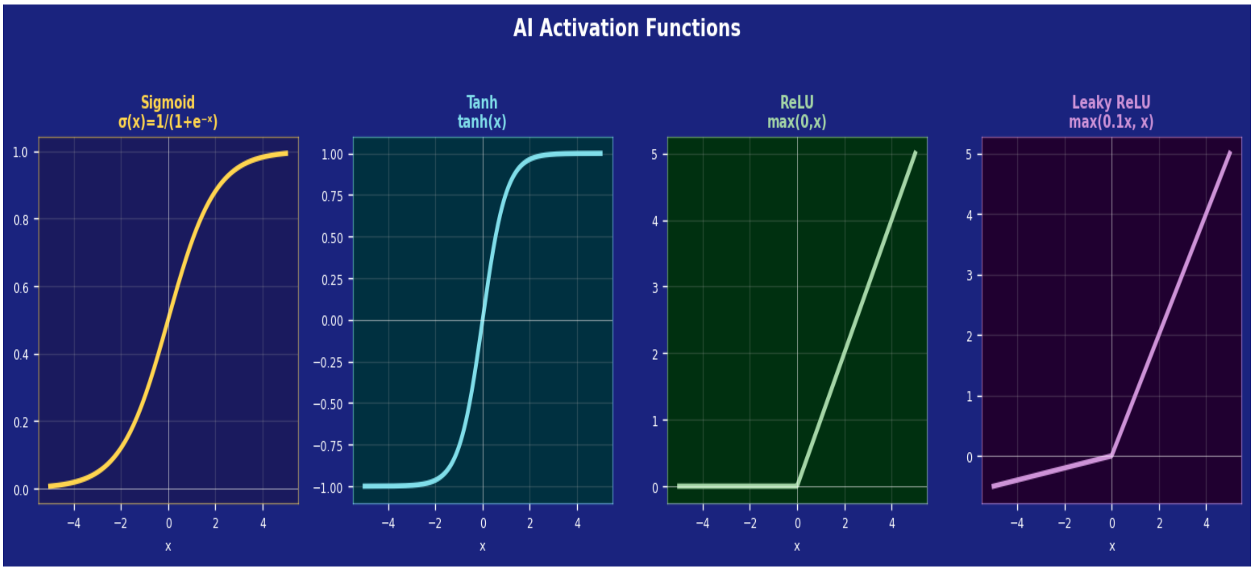

Figure: Sigmoid, Tanh, ReLU, and Leaky ReLU activation functions used in neural networks

Activation	Formula	Type	Range	Use Case
Sigmoid	σ(x) = 1/(1+e■■)	Exponential	(0, 1)	Binary classification output
Tanh	tanh(x) = (e■-e■■)/(e■+e■■)	Exponential	(-1, 1)	Hidden layers, RNNs
ReLU	f(x) = max(0, x)	Piecewise Linear	[0, ∞)	Hidden layers, CNNs
Leaky ReLU	f(x) = max(αx, x)	Piecewise Linear	■	Avoids dying ReLU
Softmax	σ■ = e■■/Σe■■	Exponential	(0,1), sum=1	Multi-class output
Linear	f(x) = x	Linear	■	Regression output layer
GELU	x·Φ(x)	Mixed	■	Transformers (BERT, GPT)

5.1 Mathematical Connections to Function Types

■ Sigmoid ↔ Exponential	σ(x) = 1/(1+e■■) uses exponential function. Smooth, differentiable, maps all reals to (0,1). Gradient: σ(x)(1-σ(x)) — important for backpropagation.
■ ReLU ↔ Piecewise Linear	f(x) = max(0,x) is a piecewise linear function. Computationally efficient; avoids vanishing gradient for positive inputs. Used in 90%+ of modern deep nets.
■ Softmax ↔ Exponential + Normalisation	Generalises sigmoid to multiple classes. Output vector sums to 1 (probability distribution). Uses exponential growth then normalises.

■ **Tanh ↔ Exponential (Shifted Sigmoid)**	$\tanh(x) = 2\sigma(2x) - 1$. Zero-centred (unlike sigmoid), beneficial for gradient flow. Range (-1,1) helps with vanishing gradient problem.
■ **Limits & Continuity**	All activation functions used in backpropagation must be continuous and (almost everywhere) differentiable. ReLU is not differentiable at x=0 but still works in practice.

6. AI Applications: Growth Models

Growth models using exponential and logarithmic functions appear throughout AI, machine learning, and data science — from learning curves to model scaling laws.

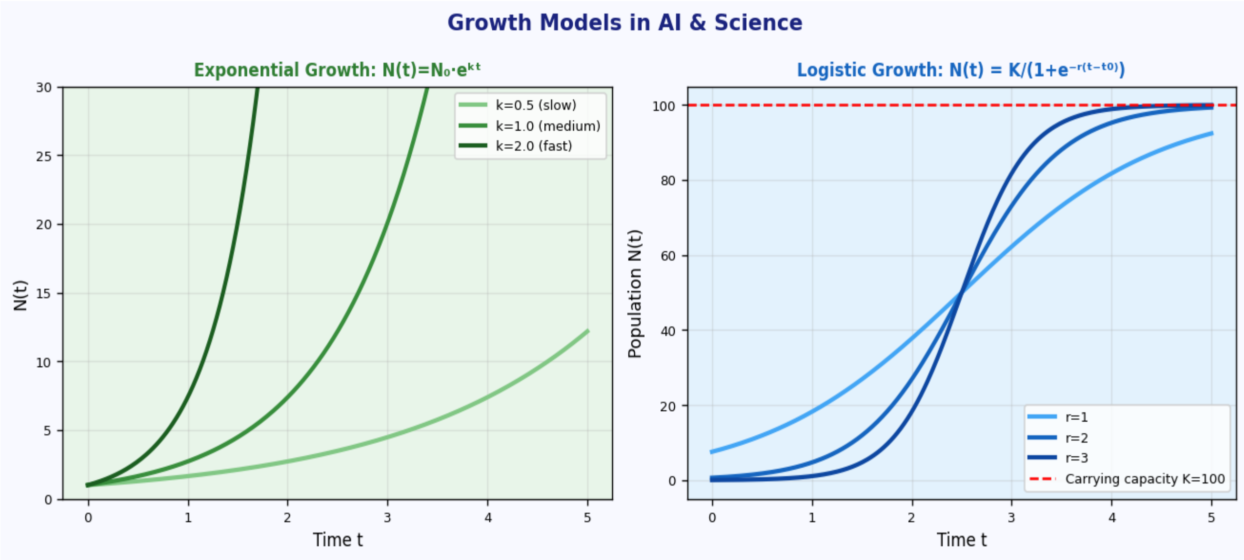

Figure: Exponential growth (various rates) and Logistic growth (sigmoid curve) used in population and AI models

6.1 Exponential Growth — Neural Scaling Laws

N(t) = N■ · e^(kt) — Continuous exponential growth

N(t) = N■ · (1+r)^t — Discrete growth (r = growth rate)

k > 0: growth | k < 0: decay | k = 0: constant

■ **Model Parameter Scaling**	The number of parameters in large language models (GPT-4, Gemini) grows exponentially with compute budget. OpenAI's scaling laws: Loss ∝ N^(-α) where N = parameters.
■ **Learning Rate Decay**	In training, learning rate often decays exponentially: η(t) = η■ · e^(-λt). Ensures fine-grained convergence at later training stages.
■ **Data Growth**	Training dataset sizes and compute (in FLOPs) have grown exponentially year over year — the foundation of the AI revolution. Doubling time ≈ 3-4 months for frontier models.
■ **Token Prediction Error**	Cross-entropy loss in language models follows a power-law (special case of exponential) as training compute scales: L(C) = (C_c/C)^(α_c).

6.2 Logistic Growth — Adoption Curves

$N(t) = K / (1 + A \cdot e^{(-rt)})$

K = carrying capacity (max saturation), r = growth rate, A = offset constant

When N is small: exponential growth. As N → K: growth slows (S-curve)

■ Technology Adoption	Sigmoid curve perfectly models AI tool adoption: slow initial uptake, exponential growth, saturation at maximum market. ChatGPT adoption followed classic S-curve.
■ Sigmoid = Logistic Function!	The sigmoid activation function σ(x) = 1/(1+e■■) IS the logistic function with K=1, r=1. Neural networks literally use growth models as activations.
■ Epidemic / Spread Models	SIR models in epidemiology (and viral content spread in social media) use logistic-type equations. Relevant for modelling information diffusion in AI systems.

6.3 Logarithmic Growth — Diminishing Returns

$f(x) = a \cdot \log_b(x) + c$

Models situations where gains slow down as input grows — diminishing returns

■ Training Loss Curves	As training progresses, loss often decreases logarithmically — initial rapid improvement, then slowly diminishing gains. More data helps, but with diminishing returns.
■ Information Theory	Entropy H = -Σ p(x)·log■(p(x)) uses logarithm. Cross-entropy loss (used to train most neural networks) is directly based on logarithmic functions.
■ Perplexity	Perplexity in language models: PP(W) = 2^H(W) = 2^(-1/N·Σlog P(w■)). Logarithm converts product of probabilities to summable values — numerical stability.

7. Key Formulae Summary

Topic	Formula / Rule	Notes
Linear Function	**f(x) = mx + b**	slope m, intercept b
Slope	**m = (y■-y■)/(x■-x■)**	rate of change
Quadratic	**f(x) = ax^2+bx+c**	parabola
Quadratic Formula	**x = [-b ± √(b^2-4ac)] / 2a**	roots of quadratic
Exponential	**f(x) = a·b■ or a·e■■**	b>1: growth, 0<b<1: decay
Log Definition	**log_b(x)=y ■ b■=x**	inverse of exponential
Log Product	**log(MN)=log M + log N**	very important!
Log Quotient	**log(M/N)=log M - log N**	subtraction rule
Log Power	**log(M■) = p·log M**	brings exponent down
Change of Base	**log_b(x) = ln(x)/ln(b)**	convert between bases
Limit Definition	**lim_{x→a} f(x) = L**	approaching, not at a
Continuity	**f cont. at a: f(a)=lim f(x)**	all 3 conditions must hold
Vertical Shift	**f(x) ± k**	up for +k, down for -k
Horizontal Shift	**f(x ■ h)**	right for -h, left for +h
Sigmoid (AI)	**σ(x) = 1/(1+e■■)**	logistic function
ReLU (AI)	**f(x) = max(0, x)**	piecewise linear
Logistic Growth	**N(t) = K/(1+A·e■■■)**	S-curve, carrying capacity K
Exponential Growth	**N(t) = N■·e■■**	continuous growth model
Cross-Entropy	**L = -Σ y·log(■)**	loss function in AI/ML

8. Practice Questions

Part A — Sets and Functions

Q1. Define a function and distinguish between domain, codomain, and range with examples. [4 marks]

Q2. Let A = {1,2,3} and B = {a,b}. List all elements of A × B. How many functions f: A → B exist? [4 marks]

Q3. Determine whether $f(x) = 2x+1$ is injective, surjective, and bijective when f: ■ → ■. [5 marks]

Q4. If $f(x) = x^2$ and $g(x) = 2x+1$, find (f■g)(x) and (g■f)(x). Are they equal? [4 marks]

Part B — Types of Functions

Q1. Find the equation of a line passing through (2,3) and (5,9). Write in slope-intercept form. [3 marks]

Q2. Solve $2x^2 - 5x + 3 = 0$ using the quadratic formula. Classify the roots. [4 marks]

Q3. Solve the equation: 3·2■ = 96. Express your answer exactly and as a decimal. [4 marks]

Q4. Simplify: log■(81) + log■(1/3) - log■(9). Use logarithm laws to show all steps. [5 marks]

Q5. If population P(t) = 1000·e^(0.05t), find the population after 10 years and the doubling time. [5 marks]

Part C — Graphs and Transformations

Q1. Describe the transformations to obtain $y = -2(x+3)^2 + 5$ from $y = x^2$. Sketch both. [6 marks]

Q2. The graph of f(x) = log(x) is transformed to g(x) = log(2x-4)+3. State domain of g(x) and all transformations applied. [5 marks]

Q3. Write the equation of f(x) = e■ after: (i) shift left 2, (ii) reflect over x-axis, (iii) shift up 4. [4 marks]

Part D — Limits and Continuity

Q1. Evaluate: $\lim_{x\to3} (x^2 - 9)/(x - 3)$. Explain why direct substitution fails. [4 marks]

Q2. Determine if $f(x) = (x^2-1)/(x-1)$ is continuous at x = 1. If not, classify the discontinuity. [5 marks]

Q3. Find $\lim_{x\to0} (\sin x)/x$ using the Squeeze Theorem. State the theorem clearly. [6 marks]

Q4. A function f(x) is defined piecewise: $f(x) = 2x+1$ for $x < 2$, and $f(x) = x^2-1$ for $x \geq 2$. Is f continuous at $x = 2$? [5 marks]

Part E — AI Applications (Applied)

Q1. Explain why the sigmoid function σ(x) = 1/(1+e■■) is used as an activation function, connecting it to the logistic growth model. [6 marks]

Q2. A neural network uses ReLU: $f(x) = \max(0,x)$. Is this function continuous? Is it differentiable everywhere? Explain with limits. [5 marks]

Q3. The cross-entropy loss is $L = -[y \cdot \log(p) + (1-y) \cdot \log(1-p)]$. If $y=1$ and $p=0.9$, compute L. What happens as $p \to 0$? [5 marks]

Q4. An AI model's parameter count N follows N(t) = 10■·e^(1.5t) (t in years). After how many years does N exceed 10^{12}? Use logarithms. [6 marks]

■ ***Study Tips:*** *Always sketch functions before working with them. For limits, try direct substitution first — if you get 0/0, factorise or rationalise. For AI applications, connect every activation function back to its mathematical function type.*